The Output Gap Does Not Exist

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Abstract

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Paragrafbudskaber fra nedenstående

% 1. No single output gap exists + Any reported gap is informative only if it states which banchmark it uses

% 2. Existing measures usually presume one latent cycle.

% 3. We build a compact multivariate UCM that separates the two gaps

% 4. Application??

% 5. Summary

Forslag til paragrafbudskaber

% 1. No single output gap exists.

% 2. Assuming only a single gap is bad

% 3. We operationalize how to distinguish several output gaps in one framework. Our approach is GENERAL and applicable to any kind of frictions and distortions ...

% 4. The key in our approach is XX. Consider for *example* the basic modern business cycle view with sticky prices, monopolistic competition and cost-push shocks ... In this view, labor hours identify deviations in output away from the socially efficient level while inflation and inflation expectations identify deviations in output away from the level under flexible prices (i teorien, og det er nok her! ikke behov for at mudre billedet atp).

% X. Applications (kan skrives senere)

% Y. Flere paragraffer med litteraturgennemgang - kan skrives løbende.

1 Introduction

There is no single output gap. Two distinct counterfactuals matter for policy: the *flex-price (natural) allocation*, which removes nominal rigidities and governs inflation pressure, and the *efficient allocation*, which removes *all* distortions and governs efficient use of resources. Deviations from each define different gaps. When the two benchmarks coincide, stabilizing inflation and closing real slack align; when they diverge, policy faces a trade-off. Any reported "gap" is informative only if it states which benchmark it uses.

Existing measures usually presume one latent cycle. Univariate trend–cycle decompositions call every stationary fluctuation "inefficient". Multivariate filters mix nominal and real indicators yet still compress demand and supply forces into a single number. In episodes with simultaneous demand shortfalls and cost–push shocks, these approaches conflate inflationary pressure with real slack, leaving policymakers unsure which frictions bind, which trade offs they face and which actions to take.

We build a compact multivariate unobserved-components model that separates the two gaps while staying close to textbook New Keynesian (NK) logic. Okun-type restrictions identify an *efficient gap* from the comovement of output, employment, and unemployment. A semi-structural Phillips curve identifies a *flex-price (inflation) gap* from inflation and expectations. The states and parameters map directly to the NK framework, keeping interpretation transparent.

Applied to standard macro data series and survey expectations, the two gaps differ in level, persistence, and timing. During cost–push episodes, the inflation gap is positive even if the efficient gap is near zero; in demand-driven slumps, the series move together. Joint estimation sharpens the Phillips-curve slope relative to single-gap models and yields a time-varying wedge between efficient and flex-price output that summarizes cost–push pressure. Forecast performance matches larger, less structured methods while producing policy-relevant objects.

The takeaway is simple: clarity about *which* gap that is being referred to is needed. Distinguishing nominal from real frictions restores structural meaning to measurement, clarifies when divine coincidence fails, and improves both identification and interpretation guiding policy makers and highlighting policy trade-offs.

2 The Empirical Two–Gap Model

We develop a compact multivariate unobserved–components (UC) model that separates two states: (i) the *efficient gap* g_t^e (real misallocation relative to the efficient allocation) and (ii) the *cost–push wedge* g_t^π (the common nominal pressure on inflation and expectations that is *not* determined by real activity). The flex-price (natural) gap g_t^f is then a *derived* object obtained from these states.

2.1 Notation and overview

Let the observed vector be

$$\mathbf{y}_t^{obs} = \mathrm{Diag}\left(\frac{1}{\boldsymbol{\sigma}}\right)\begin{pmatrix} y_t \\ e_t \\ u_t \\ \pi_t \\ F_t^{uom}\pi_{t+4} \\ F_t^{spf}\pi_{t+4} \end{pmatrix},$$

where y_t is (log) output, e_t (log) employment, u_t unemployment, π_t quarterly inflation (annualized), and $F_t^{uom}\pi_{t+4}$, $F_t^{spf}\pi_{t+4}$ are one-year-ahead survey inflation expectations.

The common state vector stacks two stochastic cycles—for the efficient gap and the cost–push wedge—plus an inflation anchor captured by the long-term inflation trend:

$$\mathbf{s}_t = \begin{pmatrix} g_t^e \\ g_t^{e*} \\ g_t^\pi \\ g_t^{\pi*} \\ \mu_t^\pi \end{pmatrix}.$$

Each (g_t, g_t^*) pair is a Harvey (1985) style stochastic cycle. μ_t^π is a random-walk long-run inflation anchor.

Idiosyncratic (series-specific) components include stochastic cycles Ψ_t^j and random-walk trends μ_t^j (with drifts in y and e):

$$\boldsymbol{\Psi}_t = \begin{pmatrix} \Psi_t^y \\ \Psi_t^e \\ \Psi_t^u \\ \Psi_t^\pi \\ \Psi_t^{uom} \\ \Psi_t^{spf} \end{pmatrix}, \qquad \boldsymbol{\mu}_t = \begin{pmatrix} \mu_t^y \\ \mu_t^e \\ \mu_t^u \\ 0 \\ \mu_t^{uom} \\ \mu_t^{spf} \end{pmatrix}.$$

2.2 Observation equations

The observation equations are

$$\mathbf{y}_t^{obs} = \mathrm{Diag}\left(\frac{1}{\boldsymbol{\sigma}}\right)\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \delta_{e,1} & \delta_{e,2} & 0 & 0 & 0 \\ \delta_{u,1} & \delta_{u,2} & 0 & 0 & 0 \\ (\delta_{E,1}+\kappa) & (\delta_{E,2}+\delta) & (\gamma_{E,1}+1) & (\gamma_{E,2}+\gamma) & 1 \\ \delta_{E,1} & \delta_{E,2} & \gamma_{E,1} & \gamma_{E,2} & 1 \\ \delta_{E,1} & \delta_{E,2} & \gamma_{E,1} & \gamma_{E,2} & 1 \end{pmatrix}}_{\equiv \mathbf{Z}}\mathbf{s}_t \;+\; \boldsymbol{\Psi}_t \;+\; \boldsymbol{\mu}_t. \quad (1)$$

Model Interpretation.

- **Real block (Okun-type restrictions).** y_t, e_t, u_t load only on the efficient cycle (g_t^e, g_t^{e*}). This pins down g_t^e from real comovement, separating real slack from nominal pressure.

- **Nominal block and the *cost–push wedge* g_t^π.** Inflation π_t and survey expectations load on $(g_t^\pi, g_t^{\pi*})$ and the anchor μ_t^π. We normalize g_t^π as the *common* component entering both inflation and expectations. Economically, g_t^π is the *cost–push wedge*: the part of nominal pressure co-moving across π_t and its expectations that is not explained by real activity.

- **NK slope on real activity.** The coefficients (κ, δ) augment the loading of (g_t^e, g_t^{e*}) in the inflation row, capturing the (semi-structural) NKPC slope of inflation with respect to real activity.

- **Idiosyncratic components.** Ψ_t^j absorb series-specific stationary residuals and μ_t^j allow permanent drifts. Inflation is level-anchored by μ_t^π, hence no separate trend for π_t.

2.3 State transition equations

Common cycles. Each cycle follows a stochastic cycle:

$$\begin{pmatrix} g_t^e \\ g_t^{e*} \end{pmatrix} = \rho_e \begin{pmatrix} \cos\lambda_e & \sin\lambda_e \\ -\sin\lambda_e & \cos\lambda_e \end{pmatrix} \begin{pmatrix} g_{t-1}^e \\ g_{t-1}^{e*} \end{pmatrix} + \begin{pmatrix} \eta_t^e \\ \eta_t^{e*} \end{pmatrix}, \tag{2}$$

$$\begin{pmatrix} g_t^\pi \\ g_t^{\pi*} \end{pmatrix} = \rho_\pi \begin{pmatrix} \cos\lambda_\pi & \sin\lambda_\pi \\ -\sin\lambda_\pi & \cos\lambda_\pi \end{pmatrix} \begin{pmatrix} g_{t-1}^\pi \\ g_{t-1}^{\pi*} \end{pmatrix} + \begin{pmatrix} \eta_t^\pi \\ \eta_t^{\pi*} \end{pmatrix}. \tag{3}$$

$0 \le \rho_\cdot < 1$ governs damping and $\lambda_\cdot \in (0, \pi)$ the frequency.

Inflation anchor. Trend inflation capturing long term inflation expectations is given by a random walk as in Hasenzagl *et al.* (2022):

$$\mu_t^\pi = \mu_{t-1}^\pi + \zeta_t^\pi, \qquad \zeta_t^\pi \sim \mathcal{N}(0, \sigma_{\zeta^\pi}^2). \tag{4}$$

Idiosyncratic cycles and trends. Each Ψ_t^j is a stochastic cycle with its own (ρ_j, λ_j, Q_j). Series-specific trends evolve as

$$\mu_t^y = \mu_{t-1}^y + \delta^y + \zeta_t^y, \qquad \mu_t^e = \mu_{t-1}^e + \delta^e + \zeta_t^e, \tag{5}$$

$$\mu_t^u = \mu_{t-1}^u + \zeta_t^u, \qquad \mu_t^{uom} = \mu_{t-1}^{uom} + \zeta_t^{uom}, \qquad \mu_t^{spf} = \mu_{t-1}^{spf} + \zeta_t^{spf}, \tag{6}$$

with mutually orthogonal innovations.

2.4 Shock structure and economic interpretation

All innovations are i.i.d. Gaussian and mutually orthogonal across blocks, *except* for a contemporaneous linkage from cost–push to the efficient cycle:

$$\eta_t^e = \phi\, \eta_t^\pi + \xi_t^e, \qquad \xi_t^e \perp \eta_t^\pi, \quad \xi_t^e \sim \mathcal{N}(0, \tilde{q}_e). \tag{7}$$

Equation (7) captures that exogenous cost–push pressure (η_t^π) can trigger policy/demand responses that move g_t^e contemporaneously. The reverse direction is shut down, preserving the exogeneity of the cost–push wedge innovations.

2.5 Why this formulation? Identification and mapping to NK logic

Separation of real and nominal pressures. The measurement matrix $\mathbf{Z}$ implements three disciplining ideas: (i) Okun-type restrictions identify the *efficient gap* g_t^e from real comovement; (ii) the *cost–push wedge* g_t^π is the common nominal movement in π_t and expectations, distinct from real slack; (iii) The shock structure allows cost–push to move real activity via policy while keeping innovations orthogonal.

Mapping to the flex-price gap. Subtracting either expectations row from the inflation row in (1) removes the anchor μ^π and the baseline loadings $(\delta_{E,\cdot}, \gamma_{E,\cdot})$, yielding

$$\pi_t - F_t\pi_{t+4} \;=\; \kappa\, g_t^e \;+\; \delta\, g_t^{e*} \;+\; g_t^\pi \;+\; \gamma\, g_t^{\pi*} \;+\; \Psi_t^\Delta, \qquad \Psi_t^\Delta \equiv \Psi_t^\pi - \Psi_t^{\exp}, \tag{8}$$

where $F_t\pi_{t+4} \in \{F_t^{uom}\pi_{t+4},\, F_t^{spf}\pi_{t+4}\}$ and $\Psi_t^{\exp}$ is the corresponding survey idiosyncratic term. We define the *flex-price (natural) gap* as the structural object that collects these terms into a single output gap:

$$g_t^f \;=\; g_t^e \;+\; \frac{\delta}{\kappa}\, g_t^{e*} \;+\; \frac{1}{\kappa}\, g_t^\pi \;+\; \frac{\gamma}{\kappa}\, g_t^{\pi*} \quad\Rightarrow\quad \pi_t - F_t\pi_{t+4} \;=\; \kappa\, g_t^f \;+\; \Psi_t^\Delta. \tag{9}$$

By (9), the flex-price gap g_t^f we formulate is the gap that summarizes *current* New Keynesian inflation pressure relative to expectations: it aggregates the efficient gap and the cost–push wedge so that $\pi_t - F_t\pi_{t+4} = \kappa\, g_t^f + \Psi_t^\Delta$. Economically, g_t^f is the *within-period* gap that, given observed expectations $F_t\pi_{t+4}$, must be closed to align inflation with expectations (ignoring idiosyncratic noise). It does not, by itself, pin inflation to the anchor μ_t^π in levels; achieving that requires expectations to sit at the anchor, which in turn would follow from a *credible commitment* to keep $g_\tau^f \equiv 0$ for all future periods.

Summary. The model delivers two key states—an efficient gap g_t^e and a cost–push wedge g_t^π—that separately capture real slack and nominal pressure. The flex-price gap g_t^f is derived via the NK slope which can be troublesome if κ is weakly identified in the data or very small (which would any how render the use of output gaps for inflation stabilization pointless). We therefore focus on the directly estimated states g^e and g^π.

3 The One-Gap Formulation Problem

3.1 Analytical Argument

3.2 Monte Carlo

4 Applications of the Two-Gap Formulation

4.1 Past Policy Evaluation

4.2 Forecasting Performance

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